Bokka Madhu Durga Sai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.82
NARAYANA JUNIOR COLLEGE	Class XII (Board Of Intermediate Education AP)	2023	97.50%
NARAYANA (E/M) HIGH SCHOOL	Class X (SSC)	2021	98.16%

SCHOLASTIC ACHIEVEMENTS

Selected as a Reliance Foundation Undergraduate Scholar, among **5,000 scholars** chosen nationwide for academic merit and leadership potential.

SKILLS

Languages : C,C++, Python, SQL, MATLAB, HTML/CSS

Frameworks: Numpy, Pandas, Matplotlib, Seaborn

Tools: Git/Github

RELEVANT COURSES

Mathematics : Calculus, Linear Algebra, Differential Equations, Introduction to Probability, Introduction to Statistics, Transform Techniques, Complex Numbers.

Computer Science : Introduction to Programming.

PROJECTS

File Compression Tool | Github

- Implemented a **Huffman Coding-based file compression utility** in C++ for lossless text compression and decompression.
- Optimized encoding using **greedy algorithms, binary trees, and STL containers**, achieving efficient memory and runtime performance.
- Developed a modular CLI application featuring file parsing, binary serialization, and validation of decompressed output.

Four-bar and slider-crank simulator (course work) | Github

- Implemented a **Python simulator for four-bar and slider-crank mechanisms**, generating kinematic plots and animations to study link motion and transmission characteristics.
- Automated kinematic analysis by calculating **position, velocity, and acceleration profiles**, reducing manual computation and improving design evaluation.

Whitworth Quick-Return Mechanism Simulator | (course work)| Github

- Developed an interactive desktop application to simulate the Whitworth Quick-Return mechanism with real-time animation, three-position synthesis, and engineering visualizations.
- Implemented kinematic and dynamic analysis to compute displacement, velocity, acceleration, input torque, and power, with interactive plots and parameterized simulation.

POSITIONS OF RESPONSIBILITY

Events Coordinator | Milan 2025, IIT Hyderabad

- Coordinated the planning and execution of multiple events for IIT Hyderabad's **inter-hostel sports fest**, collaborating with cross-functional teams to ensure smooth operations and adherence to timelines.
- Managed event logistics, volunteer coordination, and on-ground execution, contributing to efficient event delivery and an enhanced participant experience.

Hospitality Coordinator | Prayas Rural Development Center

- Coordinated hospitality arrangements for guests, volunteers, and participants during community outreach and rural development programs, ensuring smooth event operations.
- Collaborated with cross-functional teams to manage accommodation, logistics, and on-site support, contributing to an organized and welcoming experience for stakeholders.

PR & Outreach Coordinator | Diesta

- Led **public relations and outreach initiatives** for an inter-branch competition, promoting events through targeted communication and increasing student participation.
- Coordinated with student representatives, organizing teams, and stakeholders to manage event publicity, participant engagement, and information dissemination.

EXTRACURRICULAR

National Service Scheme (NSS)

- Active member of the **National Service Scheme (NSS)**, participating in community service, social awareness, and volunteer initiatives.
- Contributed to outreach and community engagement activities, promoting teamwork and social responsibility.

Milan 2025 - Volleyball

- Represented my hostel as a **Volleyball Participant** in **Milan 2025**, IIT Hyderabad's inter-hostel sports fest.
- Demonstrated teamwork, discipline, and collaboration by competing in inter-hostel volleyball tournaments.